152. Maximum Product Subarray

Given the MPS that terminates one end at index i,

calculate MPS that terminates at index $i + 1$

At $i + 1$ $\begin{cases} \text{continue } i \text{ subarray} & \text{product} \cdot = 1 \\ \text{start anew} & \text{product} = i \end{cases}$

- Given that products can flip neg -> positive too

  Keep track of most negative too

$$\text{mps_pos}^{i+1} = \max \begin{cases} \text{mps_pos_prev}^{i} \cdot \text{nums}[i] \\ \text{mps_neg_prev} \cdot \text{nums}[i] \\ \text{nums}[i] \end{cases}$$

$$\text{mps_neg}^{i+1} = \max \begin{cases} \text{mps_pos_prev}^{i} \cdot \text{nums}[i] \\ \text{mps_neg_prev} \cdot \text{nums}[i] \\ \text{nums}[i] \end{cases}$$